RESILIENCE REVIEW

# Multi-region: the cost of a second AWS region

What it would take to survive the loss of a whole AWS region: run cost, build cost and time, people, and where to put it.

> 1 service has no multi-region profile (Transform): costed by their catalog category, and their availability in other regions is not verified.

SCOPE
name contains 'prod'

ACCOUNTS GOING MULTI-REGION
40 of 40

PRIMARY REGION
Europe (London) (eu-west-2)

SECOND REGION
Europe (Frankfurt) (eu-central-1) · recommended

RECOVERY TARGET
Recover within 3 hours of a region-wide event

DATA RESIDENCY
Anywhere AWS has a region

# 1 Executive summary

BOTTOM LINE

**To recover within 3 hours, a second region in Europe (Frankfurt) for 40 accounts would add roughly $2.5m to $6m a year to run (25 to 70% of their spend), cost $1.6m to $3.5m and take 5 to 11 months to build, and need 2 to 5 more people.**

> **Figures are provisional.** They use default answers. Choose the recovery target and accounts, and answer the questions in section 2.

### WHAT WE FOUND

- **Today.** 90% of regional spend runs in Europe (London); Europe (Ireland) already carries some.
- **Recommended region.** Europe (Frankfurt): 100% of spend is on services available there, about 15 ms from Europe (London), prices about +2% against it.
- **Runners-up.** Europe (Ireland): no Connect here ($20.9k/month to re-architect). US East (N. Virginia): ~80 ms further than Europe (Frankfurt).
- **Failover risk.** 5 of 40 chosen accounts run a material service that cannot meet the target as built (no live replica: restore takes hours).
- **Data residency.** AWS has no second region in the United Kingdom: any second region moves data across a border, which needs regulator and data-protection sign-off (UK GDPR; PRA SS2/21).
- **People.** About 2 to 5 extra FTE to run two regions ($400k to $800k a year).

### WHAT WE RECOMMEND

1. **Settle residency first.** Agree with the regulator which data may sit outside the country.
2. **Tier the workloads.** Not every account needs the same target: keep instant failover for the few systems whose outage is a systemic event, and 3-hour recovery for the rest.
3. **Fix the hardest services first.** Start with services that have no live replica or are missing in the second region; they set the timeline.
4. **Rehearse.** Prove failover with a scheduled region-evacuation test and keep the evidence.

Extra cost to run (a year)

**$2.5m to $6m** PROVISIONAL

25 to 70% of the chosen accounts' spend

Cost to achieve (one-off)

**$1.6m to $3.5m** PROVISIONAL

4.5 to 10 person-years

Time to achieve

**5 to 11 months** PROVISIONAL

bound by team capacity

Extra people

**2 to 5 FTE** PROVISIONAL

$400k to $800k a year

Recommended region

**Europe (Frankfurt)**

~15 ms · 100% service parity

Accounts with failover risk

**5 of 40**

a material service cannot meet the target as built

## 2 Your choices

Saved in this browser only. Team, rates and delivery answers are shared with the exit estimate.

PROVISIONAL Using default answers. Change any answer, or confirm these.

**Recovery target after a region-wide event**

**Recover within 3 hours**

Systemic risk event: data replicated continuously, a minimal footprint scaled up on failover (pilot light).

| | | | |
|---|---|---|---|
| Where may the second region be?<br>**Anywhere AWS has a ...** | Primary region<br>**Detected: Europe (Lo...** | Second region<br>**Recommended: Euro...** | AWS Support plan<br>**Enterprise** |
| People in the core cloud team<br>**10** | % of their time on this programme<br>**50** | Application engineers available (FTE)<br>**10** | Blended day rate<br>**750** |
| Currency<br>**USD** | Cost per extra FTE a year (0 = day rate × 220)<br>**0** | Standard CI/CD pipeline?<br>**Several, partially sta...** | Terraform, and is it up to date?<br>**Terraform with signifi...** |

## 3 Accounts to make multi-region

Sorted by spend. Readiness is the worst failover risk among an account's material services for the chosen recovery target; open an account to see each service's risk.

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **data-platform-prod**<br>561017993421 · Root / Workloads / Production | $133k | High | ▾ High 5 · Moderate 3 · Low 2<br>**High** **Redshift** $62k/mo · No live replica: restore or re-run<br>No live cross-region replica. Copy snapshots to the second region and restore (hours for a large warehouse), or load both regions from the pipelines.<br>**High** **Glue** $15k/mo · No live replica: restore or re-run<br>Jobs, crawlers and the Data Catalog are regional: replicate the catalog and re-run pipelines in the second region.<br>**High** **EMR** $11k/mo · No live replica: restore or re-run<br>Clusters are rebuilt in the second region from code; data must be replicated to S3 there.<br>**High** **Athena** $6.6k/mo · No live replica: restore or re-run<br>Queries need the data and Glue catalog to exist in the second region.<br>**High** **MWAA (Airflow)** $2.0k/mo · No live replica: restore or re-run<br>Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours.<br>**Moderate** **QuickSight** $5.1k/mo · Manual or scripted failover<br>Has its own failover model, often only for specific region pairs.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $3.7k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Moderate** **Firehose** $2.6k/mo · Manual or scripted failover<br>Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over.<br>**Low** **S3** $22k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.<br>**Low** **Lake Formation** $640/mo · AWS replicates and fails over<br>Pipelines must deploy to both regions; the tooling itself can stay in one. |

| Account | Spend / month | Readiness | Multi-region risks |
|---|---|---|---|
| **core-banking-prod**<br>813958344723 · Root / Workloads / Production | $122k | Moderate | ▼ Moderate 7 · Low 1<br>**Moderate** **EC2** $56k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $34k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $15k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Moderate** **FSx** $5.7k/mo · Manual or scripted failover<br>Cross-region backup copy, or SnapMirror for FSx for NetApp ONTAP; restore or break the mirror to fail over.<br>**Moderate** **Direct Connect** $4.6k/mo · Manual or scripted failover<br>Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months.<br>**Moderate** **ELB** $2.0k/mo · Manual or scripted failover<br>Load balancers are regional; Route 53 health checks switch traffic between them.<br>**Moderate** **Amazon MQ** $1.4k/mo · Manual or scripted failover<br>Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make.<br>**Low** **Backup** $3.4k/mo · AWS replicates and fails over<br>Copy backups to a vault in the second region; restore time grows with data size. |

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| **payments-core-prod**<br>534439589175 · Root / Workloads / Production | $116k | Moderate | ▾ Moderate 10 · Low 2<br>**Moderate** **EC2** $42k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $28k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $9.9k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Moderate** **MSK (Kafka)** $7.9k/mo · Manual or scripted failover<br>MSK Replicator copies topics cross-region; consumers must handle offset translation on failover.<br>**Moderate** **EKS** $4.6k/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Moderate** **ElastiCache** $4.2k/mo · Manual or scripted failover<br>Global Datastore (Redis OSS / Valkey) replicates cross-region; without it the cache starts cold after failover.<br>**Moderate** **CloudWatch** $3.6k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.<br>**Moderate** **ELB** $2.3k/mo · Manual or scripted failover<br>Load balancers are regional; Route 53 health checks switch traffic between them.<br>**Moderate** **Lambda** $1.2k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and |

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| | | | route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br><br>Moderate **KMS** $978/mo · Manual or scripted failover<br>Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt.<br><br>Low **S3** $4.5k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points.<br><br>Low **DynamoDB** $2.8k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. |

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| **cards-processing-prod**<br>966773840736 · Root / Workloads / Production | $101k | High | ▾ High 1 · Moderate 8 · Low 2<br>**High** **Cognito** $2.4k/mo · No live replica: restore or re-run<br>User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built.<br>**Moderate** **Lambda** $23k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br>**Moderate** **API Gateway** $9.8k/mo · Manual or scripted failover<br>Deploy the API in both regions behind a custom domain with Route 53 failover.<br>**Moderate** **Kinesis Streams** $8.4k/mo · Manual or scripted failover<br>No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position.<br>**Moderate** **Step Functions** $7.1k/mo · Manual or scripted failover<br>State machines are regional; running executions do not fail over and must be restarted.<br>**Moderate** **CloudWatch** $4.8k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.<br>**Moderate** **SQS** $3.2k/mo · Manual or scripted failover<br>Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover.<br>**Moderate** **EventBridge** $1.7k/mo · Manual or scripted failover<br>Global endpoints can fail event buses over between two regions; rules and targets must exist in both.<br>**Moderate** **KMS** $1.4k/mo · Manual or scripted failover<br>Keys are regional. Data encrypted with single-Region keys cannot be decrypted in the second region: use multi-Region keys and re-encrypt.<br>**Low** **DynamoDB** $35k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer- |

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| | | | wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.<br><br>Low **S3** $4.3k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

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| **digital-channels-prod**<br>978681853473 · Root / Workloads / Production | $57k | High | ▾ High 1 · Moderate 8 · Low 1<br>**High** **Cognito** $5.6k/mo · No live replica: restore or re-run<br>User pools are regional with no replication and password hashes cannot be exported: users must be migrated or re-registered, or a sync built.<br>**Moderate** **ECS** $10k/mo · Manual or scripted failover<br>Clusters and services are regional: deploy to both regions and fail over with Route 53.<br>**Moderate** **RDS / Aurora** $9.6k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EC2** $8.0k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **AppSync** $4.3k/mo · Manual or scripted failover<br>Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over.<br>**Moderate** **Pinpoint** $2.8k/mo · Manual or scripted failover<br>Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over.<br>**Moderate** **SNS** $1.8k/mo · Manual or scripted failover<br>Topics and subscriptions are regional: recreate in the second region and publish there.<br>**Moderate** **WAF** $1.5k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.<br>**Moderate** **Amplify** $1.3k/mo · Manual or scripted failover<br>Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over. |

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| | | | Low **S3** $2.5k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **ml-platform-prod**<br>217230843944 · Root / Workloads / Production | $47k | Moderate | ▼ Moderate 6 Low 2<br>Moderate **SageMaker** $19k/mo · Manual or scripted failover<br>Models and endpoints must be redeployed in the second region; training data replicated.<br>Moderate **EC2** $6.6k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>Moderate **OpenSearch** $4.7k/mo · Manual or scripted failover<br>Cross-cluster replication to a domain in the second region; re-point clients on failover.<br>Moderate **Textract** $3.0k/mo · Manual or scripted failover<br>Models, endpoints and quotas differ by region and must be provisioned in the second region.<br>Moderate **Bedrock AgentCore** $1.2k/mo · Manual or scripted failover<br>Models, endpoints and quotas differ by region and must be provisioned in the second region.<br>Moderate **Transform** $495/mo · Manual or scripted failover<br>No cross-region failover built in. Images, launch templates and capacity must exist in the second region and be started by a runbook or automation.<br>Low **Bedrock** $7.6k/mo · AWS replicates and fails over<br>Cross-region inference spreads requests across regions, but model availability differs: check every model used exists in the second region.<br>Low **S3** $4.2k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

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| **shared-network-prod**<br>263737281768 · Root / Infrastructure | $37k | Moderate | ▼ Moderate 4<br>**Moderate** **VPC** $12k/mo · Manual or scripted failover<br>VPCs, subnets, endpoints and inter-region peering or Transit Gateway must be built in the second region.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $8.3k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Moderate** **Direct Connect** $6.6k/mo · Manual or scripted failover<br>Needs connections or a Direct Connect gateway association into the second region; physical lead times are weeks to months.<br>**Moderate** **Network Firewall** $5.3k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region. |

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| **contact-centre-prod**<br>125285552493 · Root / Workloads / Production | $34k | Moderate | ▾ Moderate 5 · Low 2<br>**Moderate** **Connect** $21k/mo · Manual or scripted failover<br>Connect Global Resiliency replicates instances and numbers between specific region pairs and is enabled by AWS on request; otherwise a second instance with its own numbers.<br>**Moderate** **Transcribe** $3.4k/mo · Manual or scripted failover<br>Models, endpoints and quotas differ by region and must be provisioned in the second region.<br>**Moderate** **Lex** $3.1k/mo · Manual or scripted failover<br>Models, endpoints and quotas differ by region and must be provisioned in the second region.<br>**Moderate** **Lambda** $2.3k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br>**Moderate** **Kinesis Streams** $1.3k/mo · Manual or scripted failover<br>No native cross-region replication: producers dual-write or a consumer replicates; consumers restart at a new position.<br>**Low** **DynamoDB** $1.8k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.<br>**Low** **S3** $1.5k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

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| **fraud-analytics-prod**<br>466806226017 · Root / Workloads / Production | $20k | High | ▼ High 1 · Moderate 4 · Low 1<br>**High** **Managed Flink** $3.6k/mo · No live replica: restore or re-run<br>Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours.<br>**Moderate** **Neptune** $6.8k/mo · Manual or scripted failover<br>Global Database for cross-region replicas; failover is a promotion step.<br>**Moderate** **SageMaker** $4.3k/mo · Manual or scripted failover<br>Models and endpoints must be redeployed in the second region; training data replicated.<br>**Moderate** **Timestream** $2.0k/mo · Manual or scripted failover<br>Needs a cross-region replica. Asynchronous replication loses the last seconds of writes, and promoting the replica is a decision someone has to make.<br>**Moderate** **Lambda** $994/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br>**Low** **DynamoDB** $2.6k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. |

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| **doc-store-prod**<br>454519794043 · Root / Workloads / Production | $9.7k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $4.8k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.9k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $986/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $975/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
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| **id-verify-prod**<br>267387487076 · Root / Workloads / Production | $8.7k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $4.4k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.6k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $877/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $877/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **investments-prod**<br>514740085648 · Root / Workloads / Production | $8.5k | Moderate | ▾ Moderate 2 · Low 1<br>**Moderate** **EC2** $5.1k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $1.7k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Low** **S3** $1.7k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

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| **mobile-backend-prod**<br>434423300025 · Root / Workloads / Production | $8.0k | Moderate | ▾ Moderate 4<br>Moderate **RDS / Aurora** $2.8k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>Moderate **EC2** $2.4k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>Moderate **ECS** $1.6k/mo · Manual or scripted failover<br>Clusters and services are regional: deploy to both regions and fail over with Route 53.<br>Moderate **Lambda** $1.2k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. |
| **identity-prod**<br>189183446249 · Root / Infrastructure | $7.9k | Moderate | ▾ Moderate 3 Low 1<br>Moderate **EC2** $3.4k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>Moderate **Directory Service** $2.2k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.<br>Moderate **ACM / Private CA** $1.6k/mo · Manual or scripted failover<br>Network, keys, secrets and security tooling are regional and must be built and kept in step in the second region.<br>Low **Secrets Manager** $647/mo · AWS replicates and fails over<br>Turn on secret replication to the second region. |

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| **payroll-api-prod**<br>295250496497 · Root / Workloads / Production | $7.6k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.8k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.3k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $758/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $755/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **branch-queue-prod**<br>717640922470 · Root / Workloads / Production | $7.3k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.7k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.2k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $732/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $728/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
|---|---|---|---|
| **insurance-quotes-prod**<br>498633168856 · Root / Workloads / Production | $7.2k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.6k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.2k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $715/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $719/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **reporting-hub-prod**<br>244219422241 · Root / Workloads / Production | $7.1k | Moderate | ▾ Moderate 2 · Low 1<br>**Moderate** **EC2** $4.3k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $1.4k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Low** **S3** $1.4k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **merchant-portal-prod**<br>386906790760 · Root / Workloads / Production | $7.0k | Moderate | ▾ Moderate 4<br>Moderate **RDS / Aurora** $2.5k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>Moderate **EC2** $2.1k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>Moderate **ECS** $1.4k/mo · Manual or scripted failover<br>Clusters and services are regional: deploy to both regions and fail over with Route 53.<br>Moderate **Lambda** $1.1k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **disputes-prod**<br>844781801768 · Root / Workloads / Production | $6.9k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.5k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $2.1k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $685/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $681/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **card-controls-prod**<br>170174047247 · Root / Workloads / Production | $6.4k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.2k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.9k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $642/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $638/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
|---|---|---|---|
| **pricing-prod**<br>322790187289 · Root / Workloads / Production | $6.1k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $3.0k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.8k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $613/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $609/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **collections-prod**<br>110356959844 · Root / Workloads / Production | $6.0k | Moderate | ▾ Moderate 2 · Low 1<br>**Moderate** **EC2** $3.6k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $1.2k/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Low** **S3** $1.2k/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **audit-trail-prod**<br>252831874844 · Root / Workloads / Production | $5.4k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $2.7k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.6k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $533/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $537/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **loans-origination-prod**<br>369887811678 · Root / Workloads / Production | $5.3k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $2.7k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.6k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $529/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $530/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **sms-gateway-prod**<br>698441060454 · Root / Workloads / Production | $4.7k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $2.4k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.4k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $470/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $471/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
|---|---|---|---|
| **kyc-checks-prod**<br>706271427442 · Root / Workloads / Production | $4.5k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $2.3k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.3k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $450/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $453/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **atm-switch-prod**<br>268805426379 · Root / Workloads / Production | $4.5k | Moderate | ▾ Moderate 2 · Low 1<br>**Moderate** **EC2** $2.7k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $896/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Low** **S3** $893/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **legacy-batch-prod**<br>107537456074 · Root / Workloads / Production | $4.2k | High | ▾ High 1 · Moderate 2 · Low 1<br>**High** **Data Pipeline** $327/mo · No live replica: restore or re-run<br>Data platforms are built single-region. Datasets, catalogs and jobs must be copied and pipelines re-run; recovery takes hours.<br>**Moderate** **EC2** $2.6k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **SWF** $712/mo · Manual or scripted failover<br>Regional endpoints. Deploy to both regions and re-route traffic; in-flight messages and running executions do not fail over.<br>**Low** **S3** $536/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **search-prod**<br>899894212661 · Root / Workloads / Production | $4.0k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **Lambda** $1.4k/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br>**Moderate** **API Gateway** $793/mo · Manual or scripted failover<br>Deploy the API in both regions behind a custom domain with Route 53 failover.<br>**Moderate** **SQS** $396/mo · Manual or scripted failover<br>Queues are regional and messages are not replicated: in-flight messages are lost or delayed on failover.<br>**Low** **DynamoDB** $1.4k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **geo-services-prod**<br>154224630493 · Root / Workloads / Production | $4.0k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $2.0k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.2k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $401/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $397/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **chatbot-prod**<br>925052687979 · Root / Workloads / Production | $3.4k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $1.7k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $1.0k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $343/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $342/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **statements-prod**<br>672004302296 · Root / Workloads / Production | $2.9k | Moderate | ▾ Moderate 4<br>**Moderate** **RDS / Aurora** $1.0k/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EC2** $881/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **ECS** $591/mo · Manual or scripted failover<br>Clusters and services are regional: deploy to both regions and fail over with Route 53.<br>**Moderate** **Lambda** $438/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. |

| Account | Spend / month | Readiness | Multi-region risks |
|---|---|---|---|
| **treasury-reporting-prod**<br>499647302627 · Root / Workloads / Production | $2.8k | Moderate | ▾ Moderate 3 · Low 2<br>**Moderate** **QuickSight** $781/mo · Manual or scripted failover<br>Has its own failover model, often only for specific region pairs.<br>**Moderate** **Lambda** $533/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them.<br>**Moderate** **Step Functions** $287/mo · Manual or scripted failover<br>State machines are regional; running executions do not fail over and must be restarted.<br>**Low** **DynamoDB** $1.0k/mo · AWS replicates and fails over<br>Global tables replicate both ways and both regions take writes; conflicting writes resolve last-writer-wins. A new replica backfills up to 10 TB per region per day unless AWS raises the quota. Import from S3 takes only 1 TB of source data at a time in Europe (Frankfurt) (15 TB only in us-east-1, us-west-2 and eu-west-1): rebuilding a large table there from an S3 export must be batched and takes longer.<br>**Low** **S3** $223/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **notifications-prod**<br>117309034122 · Root / Workloads / Production | $2.6k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $1.3k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $782/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $260/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $263/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **partner-api-prod**<br>283827598105 · Root / Workloads / Production | $2.2k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $1.1k/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $664/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $222/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $223/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **onboarding-prod**<br>149964263026 · Root / Workloads / Production | $2.1k | Moderate | ▾ Moderate 4<br>**Moderate** **RDS / Aurora** $722/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EC2** $627/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **ECS** $414/mo · Manual or scripted failover<br>Clusters and services are regional: deploy to both regions and fail over with Route 53.<br>**Moderate** **Lambda** $315/mo · Manual or scripted failover<br>Functions are regional: deploy to both regions and route with Route 53 or API Gateway; event sources (queues, streams) do not move with them. |

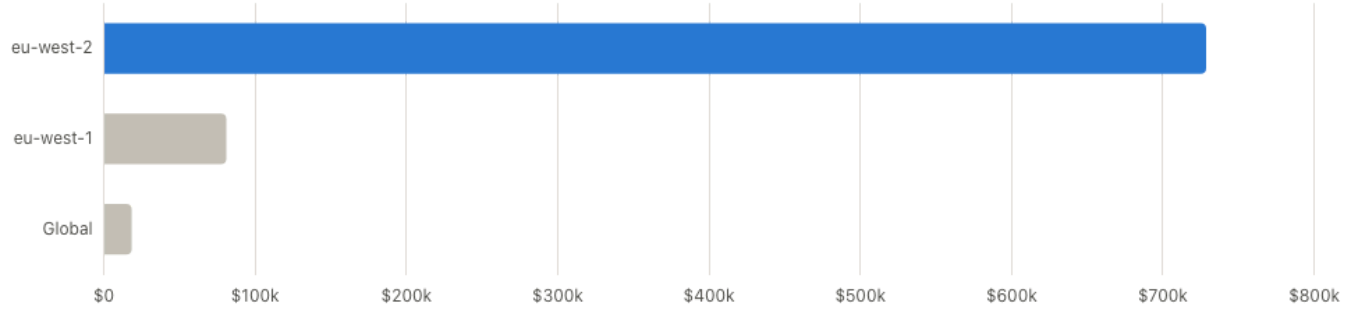
| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **rewards-prod**<br>271987579183 · Root / Workloads / Production | $1.7k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $829/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $497/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $166/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $164/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |
| **limits-prod**<br>152477260104 · Root / Workloads / Production | $1.4k | Moderate | ▾ Moderate 2 · Low 1<br>**Moderate** **EC2** $867/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **EC2 Other (EBS, NAT, transfer)** $292/mo · Manual or scripted failover<br>EBS snapshots must be copied cross-region on a schedule; NAT gateways and endpoints are rebuilt per region.<br>**Low** **S3** $289/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

| Account | Spend / month | Readiness | Multi-region risks |
| --- | --- | --- | --- |
| **fx-rates-prod**<br>753954983820 · Root / Workloads / Production | $1.0k | Moderate | ▾ Moderate 3 · Low 1<br>**Moderate** **EC2** $518/mo · Manual or scripted failover<br>Manual failover: EC2 has no cross-region failover. Copy AMIs and EBS snapshots (or use Elastic Disaster Recovery) and start capacity in the second region; capacity there is not guaranteed without reservations.<br>**Moderate** **RDS / Aurora** $313/mo · Manual or scripted failover<br>Cross-region read replica or Aurora Global Database. Promotion is a manual or scripted step (Aurora Global managed failover in about a minute); asynchronous, so a few seconds of writes can be lost.<br>**Moderate** **EKS** $105/mo · Manual or scripted failover<br>A second cluster per region (no cross-region control plane). Redeploy through GitOps; persistent volumes need their own replication.<br>**Low** **S3** $104/mo · AWS replicates and fails over<br>Cross-Region Replication (15-minute SLA with Replication Time Control). Applications must re-point, or use Multi-Region Access Points. |

## 4 Which second region

### Where the chosen accounts run today

Monthly spend by region.




### Recommended: Europe (Frankfurt)

Best balance of service parity, latency and price for the chosen accounts and recovery target.

| | |
|---|---|
| Region | Europe (Frankfurt) (eu-central-1) |
| Estimated latency from primary | ~15 ms round trip |
| Service parity (by spend) | 100.0% |
| Not available there | None |
| Prices against primary | +2% |
| Already runs there | 0.0% of regional spend |

## Candidate regions ranked by service parity, latency and price

| Region | Latency | Parity | Not available | Price | Score | Why it ranks here |
|---|---|---|---|---|---|---|
| **Europe (Frankfurt)** eu-central-1 **Recommended** | ~15 ms | 100.0% | none | +2% | 87 | Best balance of service parity, latency and price. |
| **Europe (Ireland)** eu-west-1 | ~10 ms | 97.4% | Connect | -4% | 86 | No Connect here ($20.9k/month to re-architect). 10% of spend already runs here. |
| **US East (N. Virginia)** us-east-1 | ~95 ms | 100.0% | none | -14% | 84 | ~80 ms further than Europe (Frankfurt). |
| **Canada (Central)** ca-central-1 | ~85 ms | 100.0% | none | -5% | 83 | ~70 ms further than Europe (Frankfurt). |
| **US West (Oregon)** us-west-2 | ~120 ms | 100.0% | none | -14% | 82 | ~105 ms further than Europe (Frankfurt). |
| **Europe (Stockholm)** eu-north-1 | ~25 ms | 96.7% | Connect Lex Textract | -9% | 79 | No Connect, Lex, Textract here ($26.9k/month to re-architect); ~10 ms further than Europe (Frankfurt). |
| **Europe (Paris)** eu-west-3 | ~10 ms | 97.0% | Connect Lex | +1% | 78 | No Connect, Lex here ($24.0k/month to re-architect). |
| **Europe (Spain)** eu-south-2 | ~20 ms | 96.4% | Amplify Connect Lex Transcribe | -3% | 77 | No Amplify, Connect, Lex and others here ($28.7k/month to re-architect). |
| **Asia Pacific (Seoul)** ap-northeast-2 | ~140 ms | 100.0% | none | +5% | 76 | ~125 ms further than Europe (Frankfurt); about 3% dearer than Europe (Frankfurt). |
| **US East (Ohio)** us-east-2 | ~100 ms | 97.0% | Connect Lex | -14% | 75 | No Connect, Lex here ($24.0k/month to re-architect); ~85 ms further than Europe (Frankfurt). |

## Regulation to check before data moves to Europe (Frankfurt); pointers, not legal advice

**UK GDPR (international transfers)**

Outside the UK, a transfer to Germany needs UK adequacy regulations or safeguards (the International Data Transfer Agreement or the UK Addendum to the EU clauses).

**PRA SS2/21 (outsourcing and third-party risk management)**

Notify the PRA of material outsourcing changes, which include where data is held, and include the second region in exit and stressed-exit plans.

## 5 Cost to run, and the people

Second-region infrastructure

**$2m to $4.5m** PROVISIONAL

a year, incl. replication traffic

AWS Support uplift

**$60k to $140k** PROVISIONAL

Enterprise plan, a year

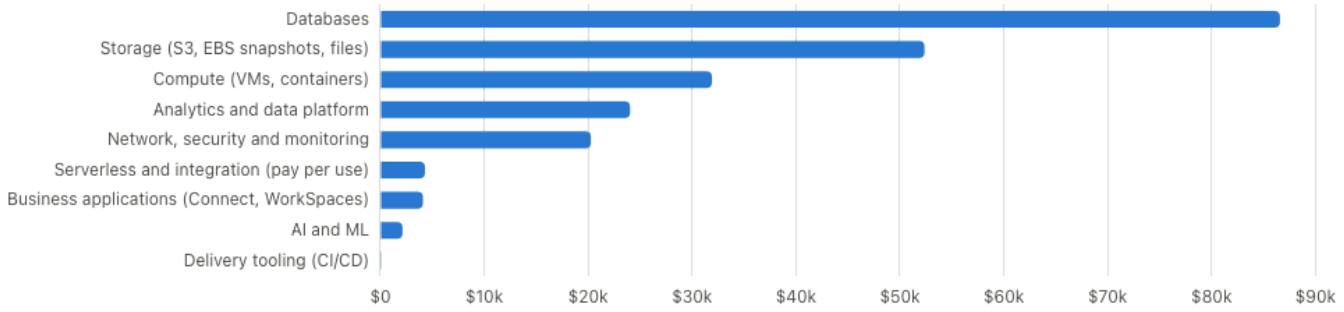
Extra people

**$400k to $800k** PROVISIONAL

2 to 5 FTE a year

### Second-region infrastructure by service class

Monthly, after the recovery target's share and regional prices.




### Extra people to run two regions

| Role | FTE (midpoint) |
|---|---|
| Platform and SRE: on-call and operations for two regions | 1.0 |
| Resilience: DR testing, failover rehearsals, regulator evidence | 1.0 |
| Workload operations across multi-region accounts | 0.4 |
| Data replication and database operations | 0.8 |
| **Total** (quoted as 2 to 5) | **3.2** |

At $165k per FTE a year.

## 6 Cost and time to achieve

Cost to achieve (one-off)

**$1.6m to $3.5m** PROVISIONAL

engineering, plus running the region while it is built

Time to achieve

**5 to 11 months** PROVISIONAL

bound by team capacity · team of 15 full-time people

Engineering effort

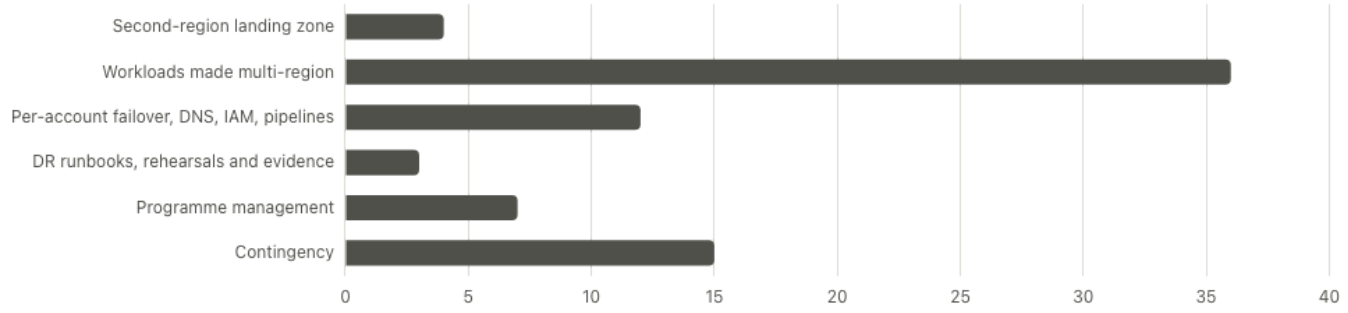
**4.5 to 10 person-years**

PROVISIONAL

40 material accounts

### Where the effort goes

Person-months, after the delivery-maturity adjustment.



## 7 Service parity in the second region

| Service | Class | Accounts | Spend / month | In eu-central-1 |
|---|---|---|---|---|
| EC2 | Compute (VMs, containers) | 33 | $188k | Available |
| RDS / Aurora | Databases | 25 | $107k | Available |
| Redshift | Analytics and data platform | 1 | $62k | Available |
| S3 | Storage (S3, EBS snapshots, files) | 31 | $55k | Available |
| DynamoDB | Databases | 6 | $44k | Available |
| EC2 Other (EBS, NAT, transfer) | Compute (VMs, containers) | 9 | $43k | Available |
| Lambda | Serverless and integration (pay per use) | 10 | $33k | Available |
| SageMaker | AI and ML | 2 | $23k | Available |
| Connect | Business applications (Connect, WorkSpaces) | 1 | $21k | Available |
| Glue | Analytics and data platform | 1 | $15k | Available |
| ECS | Compute (VMs, containers) | 5 | $14k | Available |
| EKS | Compute (VMs, containers) | 19 | $14k | Available |
| VPC | Network, security and monitoring | 1 | $12k | Available |
| Direct Connect | Network, security and monitoring | 2 | $11k | Available |
| API Gateway | Serverless and integration (pay per use) | 2 | $11k | Available |
| EMR | Analytics and data platform | 1 | $11k | Available |
| Kinesis Streams | Databases | 2 | $9.7k | Available |
| CloudWatch | Network, security and monitoring | 2 | $8.4k | Available |
| Cognito | Serverless and integration (pay per use) | 2 | $8.0k | Available |
| MSK (Kafka) | Databases | 1 | $7.9k | Available |
| Bedrock | AI and ML | 1 | $7.6k | Available |
| Step Functions | Serverless and integration (pay per use) | 2 | $7.4k | Available |
| Neptune | Databases | 1 | $6.8k | Available |

| Service | Class | Accounts | Spend / month | In eu-central-1 |
|---|---|---|---|---|
| Athena | Serverless and integration (pay per use) | 1 | $6.6k | Available |
| QuickSight | Business applications (Connect, WorkSpaces) | 2 | $5.9k | Available |
| FSx | Storage (S3, EBS snapshots, files) | 1 | $5.7k | Available |
| Network Firewall | Network, security and monitoring | 1 | $5.3k | Available |
| OpenSearch | Databases | 1 | $4.7k | Available |
| AppSync | Serverless and integration (pay per use) | 1 | $4.3k | Available |
| ELB | Network, security and monitoring | 2 | $4.2k | Available |
| ElastiCache | Databases | 1 | $4.2k | Available |
| Managed Flink | Analytics and data platform | 1 | $3.6k | Available |
| SQS | Serverless and integration (pay per use) | 2 | $3.6k | Available |
| Transcribe | AI and ML | 1 | $3.4k | Available |
| Backup | Storage (S3, EBS snapshots, files) | 1 | $3.4k | Available |
| Lex | AI and ML | 1 | $3.1k | Available |
| Textract | AI and ML | 1 | $3.0k | Available |
| Pinpoint | Serverless and integration (pay per use) | 1 | $2.8k | Available |
| Firehose | Serverless and integration (pay per use) | 1 | $2.6k | Available |
| KMS | Network, security and monitoring | 2 | $2.4k | Available |
| Directory Service | Network, security and monitoring | 1 | $2.2k | Available |
| MWAA (Airflow) | Analytics and data platform | 1 | $2.0k | Available |
| Timestream | Databases | 1 | $2.0k | Not verified |
| SNS | Serverless and integration (pay per use) | 1 | $1.8k | Available |
| EventBridge | Serverless and integration (pay per use) | 1 | $1.7k | Available |
| ACM / Private CA | Network, security and monitoring | 1 | $1.6k | Available |

| Service | Class | Accounts | Spend / month | In eu-central-1 |
|---|---|---|---|---|
| WAF | Network, security and monitoring | 1 | $1.5k | Available |
| Amazon MQ | Databases | 1 | $1.4k | Available |
| Amplify | Serverless and integration (pay per use) | 1 | $1.3k | Available |
| Bedrock AgentCore | AI and ML | 1 | $1.2k | Available |
| Transform | Compute (VMs, containers) | 1 | $495 | Not verified |
| Data Pipeline | Analytics and data platform | 1 | $327 | Not verified |

## Assumptions

- Recovery target: Recover within 3 hours of a region-wide event. Each service's spend in the chosen accounts is multiplied by its class's second-region share (data/regions.yaml): Compute (VMs, containers) 12%; Databases 45%; Storage (S3, EBS snapshots, files) 80%; Analytics and data platform 25%; Network, security and monitoring 40%; Serverless and integration (pay per use) 5%; Business applications (Connect, WorkSpaces) 15%; AI and ML 5%; Delivery tooling (CI/CD) 5%.
- Replication traffic: 6% of database, storage and analytics spend.
- Second-region prices use an indicative regional price index (EC2 on-demand list prices against us-east-1).
- AWS Support: Enterprise list-price tiers applied to the whole scanned bill plus the new region; enterprise agreements and discounts differ.
- Latency is estimated from distance (fibre ~200 km/ms, routes 1.5x the great circle): measure before committing.
- Service availability: AWS's public regional services table, version 20251113063700 (fetched 2026-10-21), corrected where it is known to be wrong (Amazon Lex available in eu-west-1). Services not listed there are 'not verified' and their effort is raised 25%.
- Region score: parity 55%, latency 20%, price 25%, plus a bonus where the estate already runs.
- People: 10% of the core cloud team (10) + 1 resilience + 0.25 per 25 material accounts + data operations scaled by data spend; at least 1.5. Cost per FTE 165,000 USD a year.
- Build: catalog-style engineer-weeks per service and account × delivery factor 1.26 (CI/CD × IaC × regulated), plus a second-region landing zone (halved where 5%+ of spend already runs there), DR programme, 12% programme management and 25% contingency.
- Duration: the longer of effort ÷ capacity (50% of the cloud team + application engineers, 80% productive) and a 20-week critical path, plus 4 weeks of failover rehearsal. Build cost includes running the half-built region meanwhile.
- Figures are ranges (−25% / +50% on cost, −20% / +40% on time), widened to round numbers.